

LLAMA: OPEN AND EFFICIENT FOUNDATION LANGUAGE MODELS EVALUATION

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Agenda

1. Evaluation Methodology

2. Benchmarks

- Categories:
 - Common Sense Reasoning
 - Close-book Question Answering
 - Reading Comprehension
 - Mathematical Reasoning
 - Code Generation
 - MMLU
- Performance Evolution During Training

3. Additional Metrics

- Instruction Finetuning (proof-of-concept)
- Harmfulness: Bias, Toxicity and Misinformation
- Environmental Damage: Carbon Footprint

4. Conclusions

5. Showcase coding the GPT model and generating text

EVALUATION METHODOLOGY



Evaluation Approach

- Language Models are Few-Shot Learners
 - Brown et al., 2020
 - [arXiv:2005.14165](https://arxiv.org/abs/2005.14165)
- Dataset based benchmarks
 - 6 categories evaluated on 17 benchmark datasets
 - 3 additional metrics
- Multiple settings
 - Zero-shot
 - Few-shot
- Two main task-types
 - Free-form generation
 - Multiple choice – ranking options by likelihood

Zero-Shot vs. Few-Shots

- Example from MMLU, a multiple-choice questionnaire dataset

Zero-Shot

Question: What is the capital of France?

Options:

A) Berlin

B) Paris

C) Rome

D) Madrid

Answer:

Few-Shots

Question: What gas do plants absorb?

Options: A) Oxygen B) Carbon dioxide C) Helium D) Nitrogen

Answer: B

Question: What is $2 + 2$?

Options: A) 3 B) 5 C) 4 D) 6

Answer: C

Question: Which animal is a mammal?

Options: A) Dolphin B) Shark C) Salmon D) Trout

Answer: A

Question: What is the capital of France?

Options:

A) Berlin

B) Paris

C) Rome

D) Madrid

Answer:

Benchmarks by Categories

1. Common Sense Reasoning

- Evaluates everyday reasoning capabilities using Cloze tasks, Winograd schemas, and multiple-choice questions.
- **Benchmarks:** BoolQ, PIQA, SIQA, HellaSwag, WinoGrande, ARC easy, ARC challenge, OpenBookQA (OBQA)

2. Closed-book Question Answering

- Tests model's ability to answer questions using only internal knowledge, without access to external resources
- **Benchmarks :** NaturalQuestions, TriviaQA

3. Reading Comprehension

- Measures understanding of text passages using middle and high school English exam questions
- **Benchmarks :** RACE-middle, RACE-high

4. Mathematical Reasoning

- Assesses capability to solve middle and high school level math problems
- **Benchmarks :** MATH, GSM8k

5. Code Generation

- Evaluates ability to generate functional Python code from natural language descriptions
- **Benchmarks :** HumanEval, MBPP

6. Massive Multitask Language Understanding (MMLU)

- A comprehensive test covering diverse domains like humanities, STEM, and social sciences
- **Benchmarks :** MMLU

Additional Metrics

■ Instruction Finetuning

- A brief finetuning on instruction dataset improves instruction-following abilities
- **Model:** LLaMA-I, evaluated on MMLU

■ Harmfulness: Bias, Toxicity & Misinformation

- Measures toxic content generation, social stereotypes, biases, and misinformation
- **Datasets:** RealToxicityPrompts (PerspectiveAPI), CrowS-Pairs, WinoGender, TruthfulQA

■ Environmental Damage by Carbon Footprint

- Estimates training energy (MWh) carbon emissions (tCO₂eq)
- Open releases reduce future environmental costs by preventing retraining

Compared Models

- **GPT-3 1.3B, 6B, 175B (OpenAI)** – The non-public industry standard baseline
- **Chinchilla 70B (DeepMind)** – Non-public model trained on 2TB of books
- **Gopher 280B (DeepMind)** – Non-public model for common reasoning tasks
- **LaMDA 137B (Google)** – General model for code generation comparison
- **Minerva 62B (Google)** – Series of PaLM models finetuned on ArXiv and Math Web Pages
- **Flan PaLM 11B, PaLM 62B, PaLM-cont 62B (Google)** – Similar to OPT-IML 30B
- **PaLM 8B, 62B, 540B (Google)** – Non-public massive scale SOTA model.
- **OPT 175B (Meta)** – Open-sourced, primary baseline for social bias and carbon footprint
- **OPT-IML 30B (Meta)** – For instruction tuning performance comparison
- **GLM 130B (Tsinghua University)** – English and Chinese pre-trained model
- **GPT-NeoX 20B, GPT-Neo (EleutherAI)** – Open-sourced models for general comparison
- **BLOOM 175B (BigScience)** – Open model, baseline for energy and carbon comparison

BENCHMARKS



Common Sense Reasoning

- **Metric:** Accuracy on multiple-choice questions, Cloze tasks, and Winograd schemas
- **Settings:** Zero-shot

		BoolQ	PIQA	SIQA	HellaSwag	WinoGrande	ARC-e	ARC-c	OBQA
GPT-3	175B	60.5	81.0	-	78.9	70.2	68.8	51.4	57.6
Gopher	280B	79.3	81.8	50.6	79.2	70.1	-	-	-
Chinchilla	70B	83.7	81.8	51.3	80.8	74.9	-	-	-
PaLM	62B	84.8	80.5	-	79.7	77.0	75.2	52.5	50.4
PaLM-cont	62B	83.9	81.4	-	80.6	77.0	-	-	-
PaLM	540B	88.0	82.3	-	83.4	81.1	76.6	53.0	53.4
LLaMA	7B	76.5	79.8	48.9	76.1	70.1	72.8	47.6	57.2
	13B	78.1	80.1	50.4	79.2	73.0	74.8	52.7	56.4
	33B	83.1	82.3	50.4	82.8	76.0	80.0	57.8	58.6
	65B	85.3	82.8	52.3	84.2	77.0	78.9	56.0	60.2

- LLaMA-65B outperforms Chinchilla-70B on almost all benchmarks
- It also surpasses PaLM-540B on most
- LLaMA-13B outperforms GPT-3 on most benchmarks despite being 10× smaller

Closed-book Question Answering

- **Metric:** Answering NatualQuestions and TriviaQA without access to external resources
- **Settings:** Zero-shot and few-shot, exact match

		0-shot	1-shot	5-shot	64-shot
GPT-3	175B	14.6	23.0	-	29.9
Gopher	280B	10.1	-	24.5	28.2
Chinchilla	70B	16.6	-	31.5	35.5
PaLM	8B	8.4	10.6	-	14.6
	62B	18.1	26.5	-	27.6
	540B	21.2	29.3	-	39.6
LLaMA	7B	16.8	18.7	22.0	26.1
	13B	20.1	23.4	28.1	31.9
	33B	24.9	28.3	32.9	36.0
	65B	23.8	31.0	35.0	39.9

		0-shot	1-shot	5-shot	64-shot
Gopher	280B	43.5	-	57.0	57.2
Chinchilla	70B	55.4	-	64.1	64.6
LLaMA	7B	50.0	53.4	56.3	57.6
	13B	56.6	60.5	63.1	64.0
	33B	65.1	67.9	69.9	70.4
	65B	68.2	71.6	72.6	73.0

- LLaMA-65B achieves state-of-the-art performance in zero-shot and few-shot settings
- LLaMA-13B competitive with GPT-3 and Chinchilla although being 5-10× smaller

Reading Comprehension

- **Metric:** Accuracy on middle and high school English reading exams (RACE)
- **Settings:** Zero-shot

		RACE-middle	RACE-high
GPT-3	175B	58.4	45.5
PaLM	8B	57.9	42.3
	62B	64.3	47.5
	540B	68.1	49.1
LLaMA	7B	61.1	46.9
	13B	61.6	47.2
	33B	64.1	48.3
	65B	67.9	51.6

- LLaMA-65B is competitive with PaLM-540B
- LLaMA-13B outperforms GPT-3 by a few percent

Mathematical Reasoning

- **Metric:** Accuracy on middle and high school math problems (MATH & GSM8k)
- **Settings:** Zero-shot, compared with majority voting

		MATH +maj1@k		GSM8k +maj1@k	
PaLM	8B	1.5	-	4.1	-
	62B	4.4	-	33.0	-
	540B	8.8	-	56.5	-
Minerva	8B	14.1	25.4	16.2	28.4
	62B	27.6	43.4	52.4	68.5
	540B	33.6	50.3	68.5	78.5
LLaMA	7B	2.9	6.9	11.0	18.1
	13B	3.9	8.8	17.8	29.3
	33B	7.1	15.2	35.6	53.1
	65B	10.6	20.5	50.9	69.7

- LLaMA-65B outperforms Minerva-62B on GSM8k
- Achieved without fine-tuning on mathematical data, unlike Minerva

Code Generation

- **Metric:** pass@1 scores, ability to generate functional Python code from description
- **Settings:** Zero-shot for HumanEval, 3-shot for MBPP

pass@	Params	HumanEval		MBPP	
		@1	@100	@1	@80
LaMDA	137B	14.0	47.3	14.8	62.4
PaLM	8B	3.6*	18.7*	5.0*	35.7*
PaLM	62B	15.9	46.3*	21.4	63.2*
PaLM-cont	62B	23.7	-	31.2	-
PaLM	540B	26.2	76.2	36.8	75.0
LLaMA	7B	10.5	36.5	17.7	56.2
	13B	15.8	52.5	22.0	64.0
	33B	21.7	70.7	30.2	73.4
	65B	23.7	79.3	37.7	76.8

- LLaMA-13B outperforms LaMDA-137B on both HumanEval and MBPP
- LLaMA-65B outperforms PaLM-62B, even with longer training
- Models in this comparison were not fine-tuned for code, as fine-tuning is out of scope

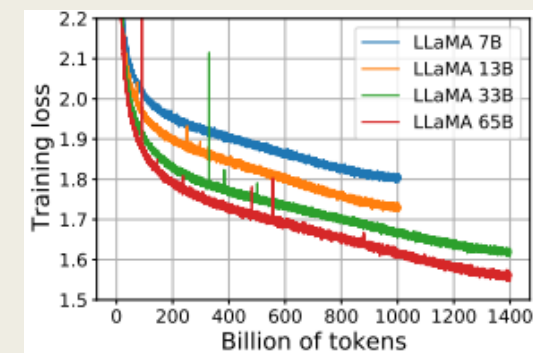
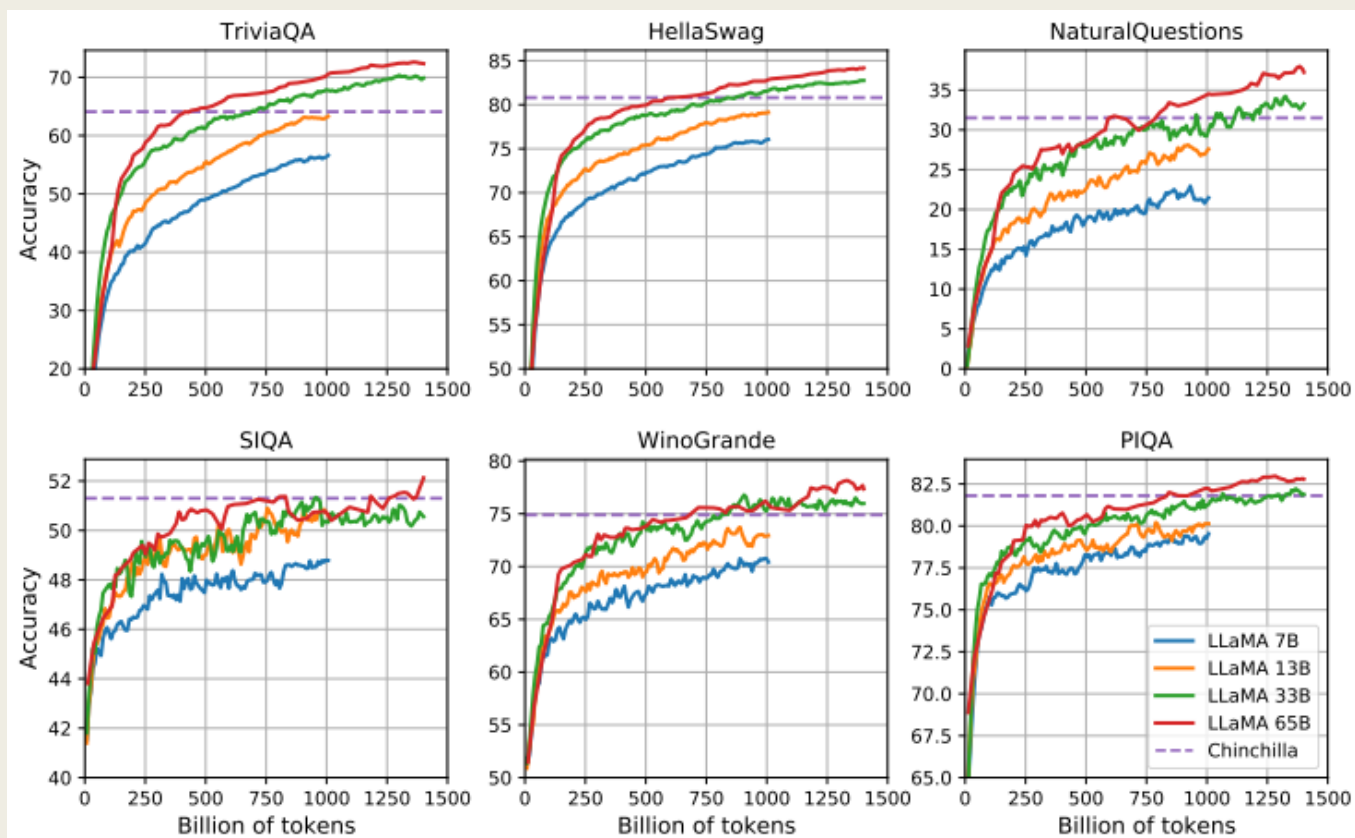
Massive Multitask Language Understanding

- **Metric:** Accuracy on multiple-choice questions across different subjects
- **Settings:** 5-shot

		Humanities	STEM	Social Sciences	Other	Average
GPT-NeoX	20B	29.8	34.9	33.7	37.7	33.6
GPT-3	175B	40.8	36.7	50.4	48.8	43.9
Gopher	280B	56.2	47.4	71.9	66.1	60.0
Chinchilla	70B	63.6	54.9	79.3	73.9	67.5
PaLM	8B	25.6	23.8	24.1	27.8	25.4
	62B	59.5	41.9	62.7	55.8	53.7
	540B	77.0	55.6	81.0	69.6	69.3
LLaMA	7B	34.0	30.5	38.3	38.1	35.1
	13B	45.0	35.8	53.8	53.3	46.9
	33B	55.8	46.0	66.7	63.4	57.8
	65B	61.8	51.7	72.9	67.4	63.4

- LLaMA-65B Falls slightly behind Chinchilla-70B and PaLM-540B
- Potentially due to limited book and ArXiv data in training (177GB vs 2TB for competitors)

Performance Evolution During Training



- Strong correlation between training perplexity and accuracy
 - As perplexity drops, benchmark scores rise.
-
- On most Question Answering and Common-Sense benchmarks, performance improves steadily as training progresses.
 - **Exceptions:**
 - The SIQA benchmark shows high variance during training, suggesting it may not be a reliable metric
 - Performance on WinoGrande does not correlate well with perplexity, with LLaMA-33B and LLaMA-65B showing similar performance despite size differences.



ADDITIONAL METRICS

Instruction Finetuning

- **Objective:** Prove that LLaMA models can be fine-tuned to follow instructions
- **Method:** Conduct a single experiment where LLaMA-65B is fine-tuned on a small instruction-tuning dataset and show that this improves its MMLU score in 5-shot setting
- **Key Results:**
 - The base LLaMA-65B model already possessed basic instruction-following abilities prior to finetuning
 - LLaMA-I (65B) reached 68.9% on MMLU, outperforming existing moderate-sized models like OPT-IML and the Flan-PaLM series
 - While strong, it remains behind the SOTA at the time (**GPT code-davinci-002**), which scored 77.4%
- **Key Takeaway:** Even a single, simple finetuning experiment yielded significant gains in instruction following and benchmark scores

OPT	30B	26.1
GLM	120B	44.8
PaLM	62B	55.1
PaLM-cont	62B	62.8
Chinchilla	70B	67.5
LLaMA	65B	63.4
OPT-IML-Max	30B	43.2
Flan-T5-XXL	11B	55.1
Flan-PaLM	62B	59.6
Flan-PaLM-cont	62B	66.1
LLaMA-I	65B	68.9

Harmfulness: Bias, Toxicity and Misinformation

- Large language models are trained on large amount of Web data
- They can reproduce and amplify **biases** found in their **training data**
- They may also generate **toxic or offensive** content
- LLaMA models are no exception
- They were evaluated on benchmarks for toxicity and stereotype detection
- These benchmarks are standard in the community but are **not sufficient to fully capture all risks associated with such models**

Harmfulness: Bias, Toxicity and Misinformation

■ RealToxicityPrompts

- Evaluate Toxicity Score (0 to 1) using **PerspectiveAPI**
- Toxicity tends to increase with model size
- Largest model is toxic when given “Respectful” prompts
- Comparisons are difficult due to different in sampling strategies and API versions

		Basic	Respectful
LLaMA	7B	0.106	0.081
	13B	0.104	0.095
	33B	0.107	0.087
	65B	0.128	0.141

■ CrowS-Pairs & WinoGender

- Evaluate model stereotypical biases
- LLaMA compares slightly favorably to GPT-3 and OPT-175B on average
- Biased in the **Religion**, followed by **Age** and **Gender** → Attributed to the CommonCrawl
- Performs better in neutral pronouns ("their/them") compared to gendered ones ("her/his")
- Performance drops in “gotcha” cases, e.g., a male nurse.

■ TruthfulQA

- Test model ability to reply **truthful** (factual) answers, and **truthful & informative** answers
- LLaMA scores **higher than GPT-3** in both “Truthful” and “Truthful & Informative”
- The absolute rate of correct answers **remains low** (prone to hallucinate)

CrowS-Pairs

- **Metric:** Preference for stereotypical sentences, measured by the perplexity of the model on two version of a sentence – one stereotypical and one neutral.
- **Settings:** Zero-shot, 9 categories.
- **Example** [[arxiv.org:2010.00133](https://arxiv.org/abs/2010.00133)]:
*Danny had to force his partner **James** to watch football with him* → 0.3
*Danny had to force his partner **Judy** to watch football with him* → 0.1 → model is stereotypical
- Overall, LLaMA compares slightly favorably to GPT-3 and OPT-175B on average.
- The model is **particularly biased in the Religion**, followed by **Age** and **Gender**.
- These biases are attributed to the CommonCrawl dataset, despite the filtering steps applied.

	LLaMA	GPT3	OPT
Gender	70.6	62.6	65.7
Religion	79.0	73.3	68.6
Race/Color	57.0	64.7	68.6
Sexual orientation	81.0	76.2	78.6
Age	70.1	64.4	67.8
Nationality	64.2	61.6	62.9
Disability	66.7	76.7	76.7
Physical appearance	77.8	74.6	76.2
Socioeconomic status	71.5	73.8	76.2
Average	66.6	67.2	69.5

Table 12: **CrowS-Pairs**. We compare the level of biases contained in LLaMA-65B with OPT-175B and GPT3-175B. Higher score indicates higher bias.

WinoGender

- **Metric:** Co-reference resolution accuracy.
- **Settings:** Zero-shot with Winograd schemas that use "her", "his", and "their" pronouns.
- **Example:**
 - *The **physician** hired the **nurse** because **he** was busy.*
 - *The **physician** hired the **nurse** because **she** was busy.*
- The model performs better when resolving **neutral pronouns** ("their/them") compared to **gendered ones** ("her/his").
- Performance drops noticeably in “gotcha” cases, when the pronoun does not match the occupation's historical majority gender
 - e.g., a male nurse.

	7B	13B	33B	65B
All	66.0	64.7	69.0	77.5
her/her/she	65.0	66.7	66.7	78.8
his/him/he	60.8	62.5	62.1	72.1
their/them/someone	72.1	65.0	78.3	81.7
her/her/she (gotcha)	64.2	65.8	61.7	75.0
his/him/he (gotcha)	55.0	55.8	55.8	63.3

Table 13: **WinoGender**. Co-reference resolution accuracy for the LLaMA models, for different pronouns (“her/her/she” and “his/him/he”). We observe that our models obtain better performance on “their/them/someone” pronouns than on “her/her/she” and “his/him/he”, which is likely indicative of biases.

TruthfulQA

- **Metric:** Percentage of:
 - “**Truthful**” answers – responses that are factually correct according to reality.
 - “**Truthful & Informative**” answers – also provide useful details to fully answer the question.
- **Settings:** Zero-shot with adversarial questions across 38 categories.
- LLaMA scores **higher than GPT-3** in both "Truthful" and "Truthful & Informative" categories.
- Limitations: The absolute rate of correct answers **remains low**.
- The model is still **prone to hallucinating incorrect answers**, indicating significant room for improvement in handling misinformation.

		Truthful	Truthful*Inf
GPT-3	1.3B	0.31	0.19
	6B	0.22	0.19
	175B	0.28	0.25
LLaMA	7B	0.33	0.29
	13B	0.47	0.41
	33B	0.52	0.48
	65B	0.57	0.53

Table 14: **TruthfulQA**. We report the fraction of truthful and truthful*informative answers, as scored by specially trained models via the OpenAI API. We follow the QA prompt style used in [Ouyang et al. \(2022\)](#), and report the performance of GPT-3 from the same paper.

Example: WinoGender

- Assume we take the following sentence from the WinoGender dataset:
 - “The **nurse** notified the **patient** that **his** shift would be ending in an hour.”
- Logically, the pronoun “his” refers to the nurse, so we infer that the nurse is male
- However, if most nurses in the model’s training data are described as female, the model may absorb this stereotype and incorrectly resolve the pronoun
- To test this, we can feed the sentence into the model and ask directly who “his” refers to
- A better approach is to compare the model’s perplexity for different candidate continuations (“the nurse” vs. “the patient”) to see which entity the model believes “his” most likely refers to.
- LLaMA handle neutral pronouns well, but make more mistakes with gendered pronouns, showing they likely have gender bias

	7B	13B	33B	65B
All	66.0	64.7	69.0	77.5
her/her/she	65.0	66.7	66.7	78.8
his/him/he	60.8	62.5	62.1	72.1
their/them/someone	72.1	65.0	78.3	81.7
her/her/she (<i>gotcha</i>)	64.2	65.8	61.7	75.0
his/him/he (<i>gotcha</i>)	55.0	55.8	55.8	63.3

Environmental Damage: Carbon Footprint

- Estimates based on Wu et al. (2022) formulas:

$$\text{Wh} = \text{GPU-h} \times (\text{GPU-power} - \text{consumption}) \times \text{PUE}$$

- Carbon emission conversion:

$$\text{tCO}_2\text{eq} = \text{MWh} \times 0.385$$

- LLaMA Training Statistics:

- Resources: 2048 A100-80GB GPUs used for approximately 5 months
- Total Energy: ~2,638 MWh
- Total Emissions: **~1,015 tCO₂eq**
 - A round-trip flight between New York and London taken 1,000 times

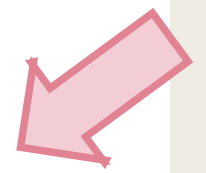
- Environmental Impact:

- Open Science: Releasing these models is intended to reduce future emissions by preventing the need for the community to retrain similar models from scratch
- Accessibility: The smaller LLaMA models are designed to run on a single GPU, significantly lowering the energy barrier for research and inference

Environmental Damage: Carbon Footprint

- The following table compares the energy use and carbon emissions from training LLaMA models with those of larger models that achieve similar performance

	GPU Type	GPU Power consumption	GPU-hours	Total power consumption	Carbon emitted (tCO ₂ eq)
OPT-175B	A100-80GB	400W	809,472	356 MWh	137
BLOOM-175B	A100-80GB	400W	1,082,880	475 MWh	183
LLaMA-7B	A100-80GB	400W	82,432	36 MWh	14
LLaMA-13B	A100-80GB	400W	135,168	59 MWh	23
LLaMA-33B	A100-80GB	400W	530,432	233 MWh	90
LLaMA-65B	A100-80GB	400W	1,022,362	449 MWh	173



- The training process of LLaMA models is expensive and environmentally harmful, especially compared to larger models
- **However, training is done only once, whereas inference is run many times**



CONCLUSIONS

Conclusions

Summary

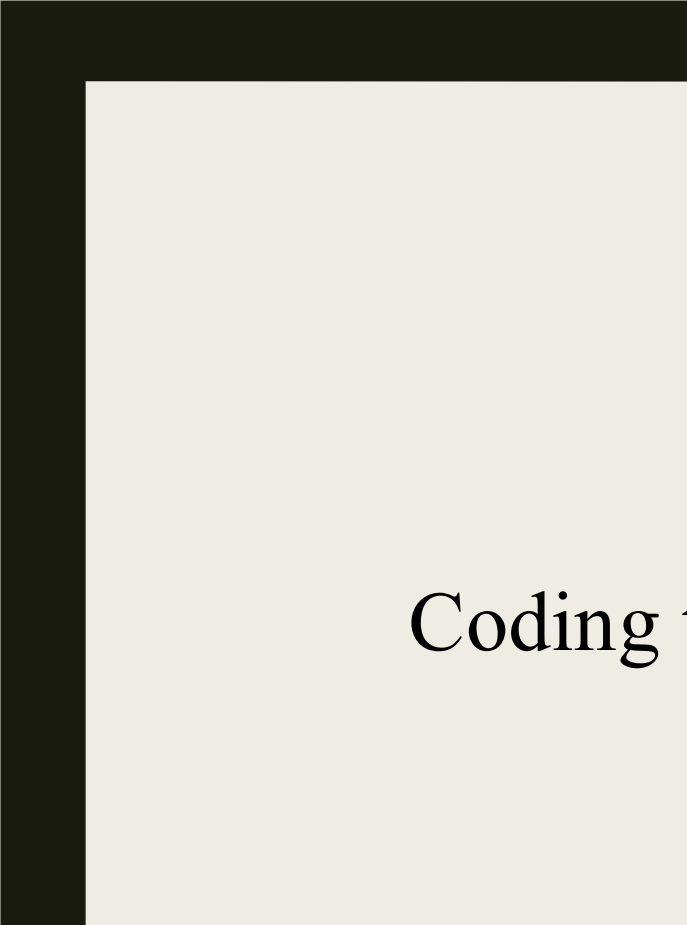
- The article presented a series of open-source language models competitive with state-of-the-art foundation models
- The authors demonstrated that **SOTA performance** can be achieved using exclusively **publicly available data**, without proprietary datasets
- Instead of continually increasing model size to improve performance, we can simply train the models for longer

Performance Highlights

- LLaMA-13B outperforms GPT-3 while being $>10x$ smaller.
- LLaMA-65B is competitive with Chinchilla-70B and PaLM-540B.

Future Directions

- Accelerate research to improve **robustness** and mitigate issues **bias and toxicity** issues
- Further investigate the **instruction finetuning** of LLaMA
- Release **larger models** trained on **larger corpora**, as performance continues to improve with scale



LLM From Scratch

Coding the GPT model & Generating text

[ch04_presentation.ipynb](#)

